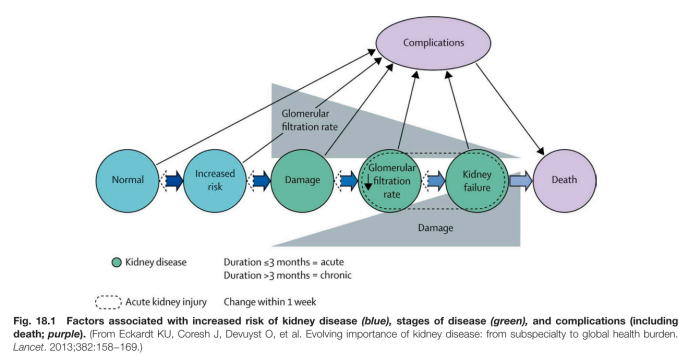
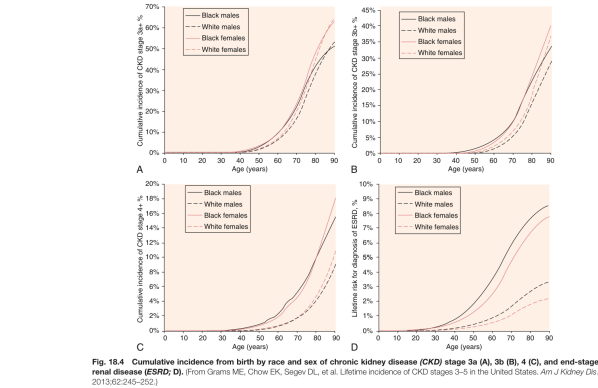
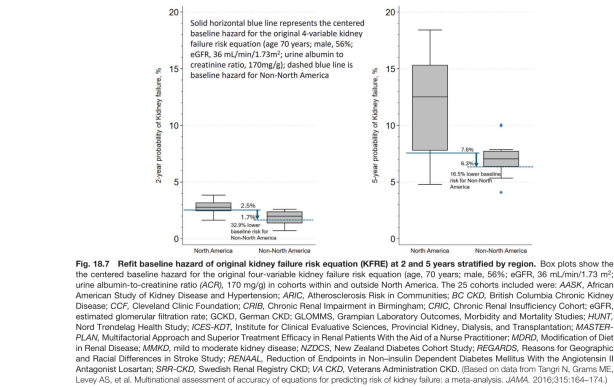




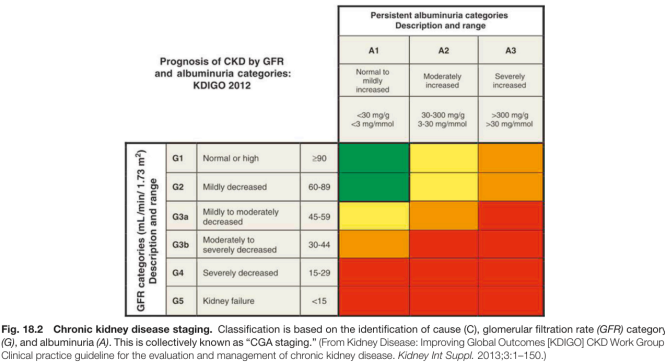
Fig_18_1



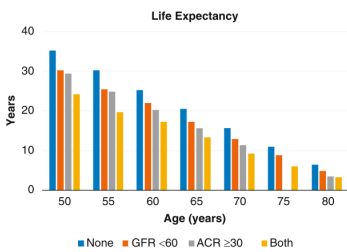
Fig_18_4



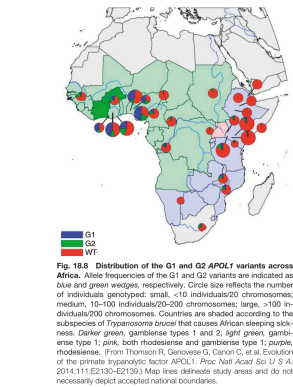
Fig_18_7



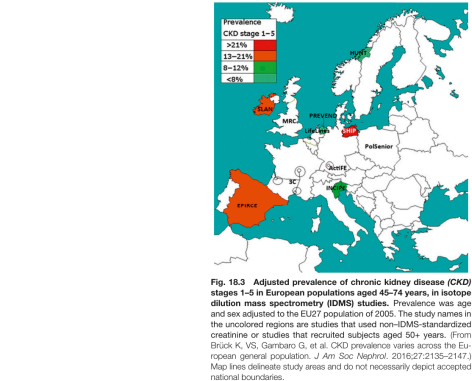
Fig_18_2



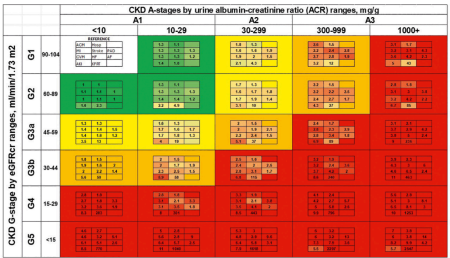
Fig_18_5



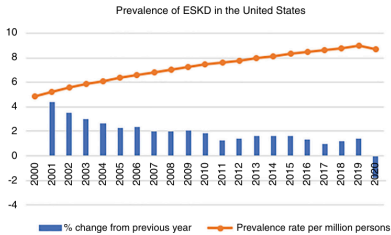
Fig_18_8



Fig_18_3



Fig_18_6



Fig_18_9

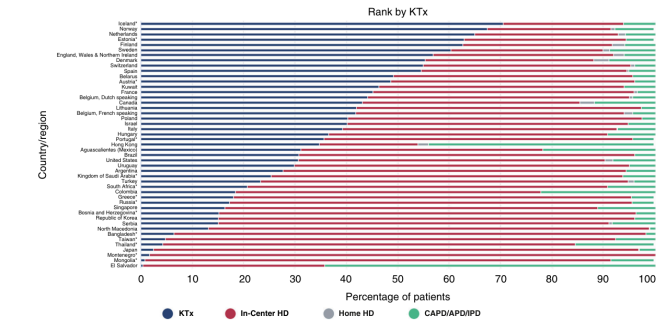


Fig. 18.10 Percentage distribution of type of kidney replacement therapy modality used by patients with end-stage kidney disease, by country, in 2020. HD, Hemodialysis. (From U.S. Renal Data System. 2022 USRDS annual data report: epidemiology of kidney disease in the United States.)

Fig_18_10

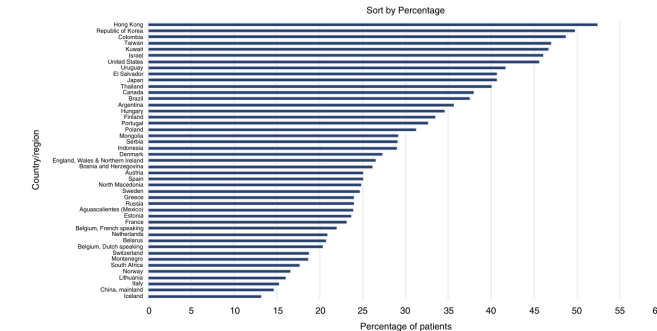


Fig. 18.13 Proportions of incident end-stage kidney disease (ESKD) due to diabetes, by country, 2020. (From U.S. Renal Data System. 2022 USRDS annual data report: epidemiology of kidney disease in the United States.)

Fig_18_13

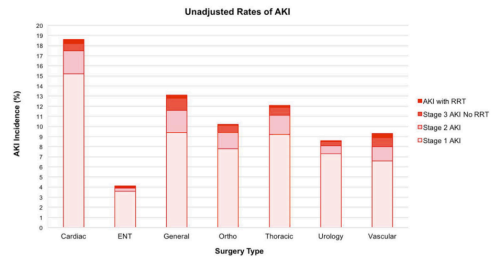


Fig. 18.16 Unadjusted rates of acute kidney injury, by surgery type and stage. AKI, Acute kidney injury; ENT, ear, nose, throat; RRT, renal replacement therapy. (Adapted from Grams ME, Sang Y, Corsh J, et al. Acute kidney injury after major surgery: a retrospective analysis of Veterans Health Administration data. *Am J Kidney Dis*. 2016;67(6):872-880.)

Fig_18_16

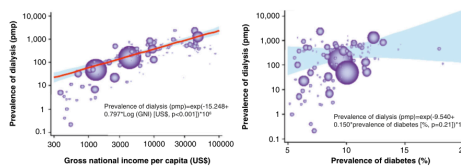


Fig. 18.11 Association between prevalence of end-stage kidney disease (ESKD) treated by dialysis in 123 countries (gross national income per capita [left], prevalence of diabetes [right]). In both graphs, China and India are represented by the largest and second largest circles, respectively. (From Lysaght T, Ninyori T, Jha V, et al. Worldwide access to treatment for end-stage kidney disease: a systematic review. *Lancet*. 2015;385:1975-1982.)

Fig_18_11

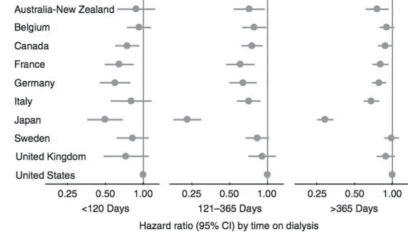


Fig. 18.14 Mortality in each DOPPS (Dialysis Outcomes and Practice Patterns Study) country versus the United States by time on hemodialysis (HD). Model adjusted for age, sex, race, and diabetes as the cause of end-stage kidney disease, stratified by study phase (N = 86,886 HD patients from DOPPS census [2002-2008]). CI, Confidence interval. (From Robinson BM, Zhang J, Morgenstern H, et al. Worldwide, mortality risk is high soon after initiation of hemodialysis. *Kidney Int*. 2014;85:158-165.)

Fig_18_14

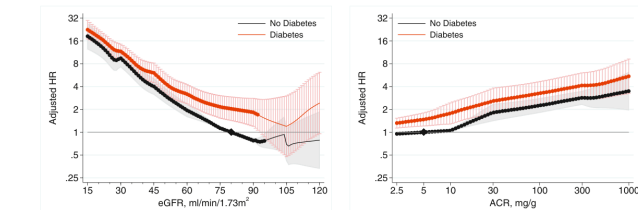


Fig. 18.17 Acute kidney injury risk by estimated glomerular filtration rate (eGFR) and urine albumin-to-creatinine ratio (ACR), stratified by diabetes status. Hazard ratios (HRs; with 95% CI [confidence interval]) of acute kidney injury, using common referent group without diabetes, according to eGFR (reference, 80 mL/min/1.73 m² without diabetes; left) and ACR (reference, 5 mg/g without diabetes; right) in individuals with and without diabetes in general population cohorts. (From James MT, Grams ME, Woodward M, et al. A meta-analysis of the association of estimated GFR, albuminuria, diabetes mellitus, and hypertension with acute kidney injury. *Am J Kidney Dis*. 2015;66:602-612.)

Fig_18_17

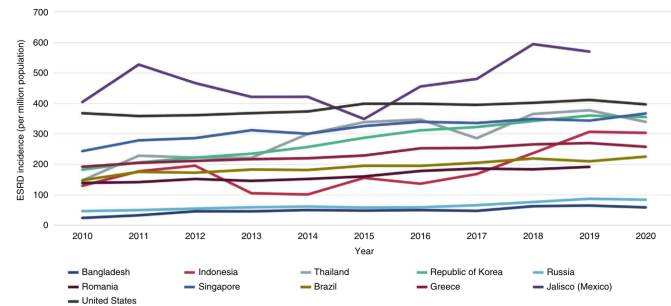


Fig. 18.12 Trends in the incidence rate of treated end-stage kidney disease (ESKD; per million population/year), by country, 2010-2020. The 10 countries with the highest percentage rise in ESKD incidence rate over the past decade, plus the United States. (From U.S. Renal Data System. 2022 USRDS Annual data report: epidemiology of kidney disease in the United States.)

Fig_18_12

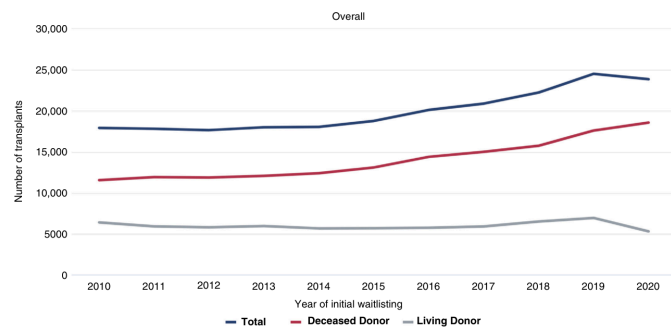


Fig. 18.15 Number of kidney transplants by donor type, 2010-2020, in patients undergoing dialysis. (Data from U.S. Renal Data System. 2022 Annual data report: atlas of chronic kidney disease and end-stage renal disease in the United States.)

Fig_18_15

Table 18.1 Definitions and Classification of Chronic Kidney Disease (CKD), Acute Kidney Disease (AKD), and Acute Kidney Injury (AKI)			
Parameter	CKD	AKD	AKI
Definition	Functional criteria GFR < 60 mL/min/1.73 m²	AKI or GFR < 60 mL/min/1.73 m² or Increase in SCr by ≥0.3 mg/dL within 7 days or Increase in SCr by ≥50% within 7 days	Increase in serum creatinine level (SCr) by ≥0.3 mg/dL within 7 days or Increase in SCr by ≥0.3 mg/dL within 7 days or Increase in SCr by ≥50% within 7 days
Structural criteria	Duration ≥3 months	Kidney damage ≥3 months	No criteria ≥3 months
Classification	None	Same as AKI	Decreased kidney perfusion, obstruction of the urinary tract, prerenal/renal kidney disease, other than acute kidney disease, and acute kidney disease
Cause	Phenomenon or disease of systemic disease and location in the kidney of pathologic evidence (glomerular, tubular, and interstitial)	Same as CKD	Not specified separately
Severity (stage)	AKI: Normal or high GFR (≥90 mL/min/1.73 m²) or AKI: Normal to mildly decreased GFR (60-89 mL/min/1.73 m²) or AKI: Moderately decreased GFR (30-59 mL/min/1.73 m²) or AKI: Severely decreased GFR (<30 mL/min/1.73 m²)	Not specified separately	Stage based on SCr or urine output 1. SCr ≥1.5 times baseline or <0.5 mL/kg/12 h or <0.5 mL/kg/24 h 2. SCr ≥2.0 times baseline or <0.5 mL/kg/12 h or <0.5 mL/kg/24 h 3. SCr ≥3.0 times baseline or <0.5 mL/kg/12 h or <0.5 mL/kg/24 h
Risk categories	AKI: Normal to mildly decreased GFR (60-89 mL/min/1.73 m²) or AKI: Moderately decreased GFR (30-59 mL/min/1.73 m²) or AKI: Severely decreased GFR (<30 mL/min/1.73 m²)	Not specified separately	Stage 1, 2, and 3
Treatment	AKI: Normal to mildly decreased GFR (60-89 mL/min/1.73 m²) or AKI: Moderately decreased GFR (30-59 mL/min/1.73 m²) or AKI: Severely decreased GFR (<30 mL/min/1.73 m²)	Not specified separately	AKI: Normal to mildly decreased GFR (60-89 mL/min/1.73 m²) or AKI: Moderately decreased GFR (30-59 mL/min/1.73 m²) or AKI: Severely decreased GFR (<30 mL/min/1.73 m²)

Table_18_1